ECEN/CSEN 123 Prelab 7

Dylan Thornburg, Austin Petersen, Madeline Follosco

II. Pre-Lab

Preparation:

• Review the MCP4811 datasheet
• Review the PAM8302 datasheet
• Review the class example code in Sine_Example_table8 from the class examples folder and understand it.
• Review the Arduino SPI library documentation and functions.
https://docs.arduino.cc/learn/communication/spi/

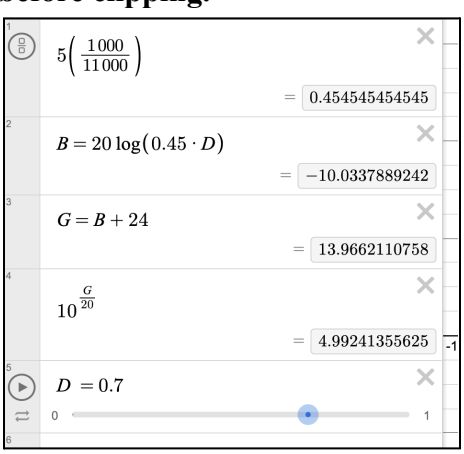
Pre-Lab Report:

• Determine the current sine-wave output frequency of the Sine_Example_table8 from the class examples folder

Printrate = 50. When millis > printrate, the loop executes and resets our millis counter (lastPrintTime). Thus we sample 20 times per second.
With a full cycle being completed every 80 samples (from 64 * 256/80), this puts our effective frequency at .25Hz.

• Explain why resistors R1 and R3 are used.

R1 and R3 form a voltage divider to divide the Arduino pin voltage from 5V to 0.45V [Vin*(1000/11000)]. This amplifier has Vdd set at 5V, so our amplifier can not amplify above 5v, so our input voltage needs to be below the max to not hit the rail and distort the signal. With a gain of 24 dB, this is a large gain that will boost our 0.45V input up to 7.16V, which will clip without any PWM, but with PWM we can have a duty cycle of up to 70% before clipping.



• Include a selfie from your planning meeting standing in front of a sign.